Rotating medium — Solution

W26 (2026) — English translation

A1 — 1.30 pts

Find the trajectory of motion $\vec{r}(t)$ for this particle. Express the answer in vector form through $\vec{v}_0$, $\vec{g}$, ε or express the dependences of the components of the vector $\vec{r}(t)$ in the Cartesian coordinate system through the components of the vectors $\vec{v}_0$, $\vec{g}$.

Newton's second law:

$$m\dot{\vec{v}} = -\varepsilon m\vec{v} + m\vec{g}.$$

Let us split the velocity into two terms:

$$\vec{v}(t) = \vec{u}(t) + \frac{\vec{g}}{\varepsilon}.$$

The equation for $\vec{u}$ is:

$$\dot{\vec{u}} = -\varepsilon\vec{u}.$$

$$\vec{v}(t) = \frac{\vec{g}}{\varepsilon} + \left(\vec{v}_0 - \frac{\vec{g}}{\varepsilon}\right)e^{-\varepsilon t}$$

Answer:

$$\vec{r}(t) = \frac{\vec{g}t}{\varepsilon} + \left(\vec{v}_0 - \frac{\vec{g}}{\varepsilon}\right)\frac{1 - e^{-\varepsilon t}}{\varepsilon}$$

A2 — 0.70 pts

How does the power of energy loss $P(t)$, which goes into heat, depend on time? Express your answer in terms of $\vec{v}_0$, $\vec{g}$, ε, t, m.

Let us write the law of energy change for the system of the particle and the medium:

$$m\left(\vec{v}, \dot{\vec{v}}\right) = m\left(\vec{v}, \vec{g}\right) - P.$$

$$P = -m\left(\vec{v}, \dot{\vec{v}} - \vec{g}\right) = \varepsilon m v^2$$

$$\vec{v}(t) = \frac{\vec{g}}{\varepsilon}\left(1 - e^{-\varepsilon t}\right) + \vec{v}_0 e^{-\varepsilon t}$$

$$v^2 = \frac{g^2\left(1 - e^{-\varepsilon t}\right)^2}{\varepsilon^2} + v_0^2 e^{-2\varepsilon t} + \frac{2\left(\vec{v}_0, \vec{g}\right)}{\varepsilon}\left(1 - e^{-\varepsilon t}\right)e^{-\varepsilon t}$$

Answer:

$$P = \varepsilon m\left(\frac{g^2\left(1 - e^{-\varepsilon t}\right)^2}{\varepsilon^2} + v_0^2 e^{-2\varepsilon t} + \frac{2\left(\vec{v}_0, \vec{g}\right)}{\varepsilon}\left(1 - e^{-\varepsilon t}\right)e^{-\varepsilon t}\right)$$

A3 — 2.00 pts

Find the loss power P, which goes into heat, a long time after the power $t \gg 1/\varepsilon$ is turned on. Express your answer using F_0, ε, m, ω.

After a long time $t \gg 1/\varepsilon$, steady motion is established, and from the isotropy of the system it is clear that the motion will occur in a circle with an angular velocity ω and a constant speed u. In this case, the angle between the velocity and force is constant and equal to α. Let us write down Newton's second law in projection onto the direction of velocity and the normal to it.

$$\begin{cases} 0 = F_0 \cos\alpha - \varepsilon m u \\ m u \omega = F_0 \sin\alpha \end{cases}$$

$$\tan\alpha = \frac{\omega}{\varepsilon}$$

$$u = \frac{F_0 \cos\alpha}{\varepsilon m} = \frac{F_0}{\varepsilon m} \frac{\varepsilon}{\sqrt{\varepsilon^2 + \omega^2}} = \frac{F_0}{m\sqrt{\varepsilon^2 + \omega^2}}$$

The law of energy change for the system is:

$$P = F_0 u \cos\alpha = \frac{F_0^2 \varepsilon}{m\left(\varepsilon^2 + \omega^2\right)}.$$

Answer:

$$P = \frac{F_0^2 \varepsilon}{m\left(\varepsilon^2 + \omega^2\right)}$$

B1 **0.50 pts**

Find the dependence $z(t)$. Express your answer in terms of r_0, ε, g.

The projection of Newton's second law onto the z axis is:

$$m\ddot{z} = -mg - \varepsilon m \dot{z}.$$

$$\ddot{z} = -g - \varepsilon \dot{z}$$

We solve this similarly to the previous part of the problem.

Answer:

$$z(t) = -\frac{gt}{\varepsilon} + \frac{g\left(1 - e^{-\varepsilon t}\right)}{\varepsilon^2}$$

B2 **1.00 pts**

Express $\ddot{x}$ and $\ddot{y}$ in terms of x, y, $\dot{x}$, $\dot{y}$, ε and ω.

Newton's second law:

$$m\ddot{\vec{r}} = m\vec{g} + \varepsilon m \left([\vec{\omega}, \vec{r}] - \dot{\vec{r}}\right).$$

In projection onto the x and y axes:

Answer:

$$\begin{cases} \ddot{x} = \varepsilon\left(-\omega y - \dot{x}\right) \\ \ddot{y} = \varepsilon\left(\omega x - \dot{y}\right) \end{cases}$$

B3 **1.00 pts**

Using the solution (x_1, y_1), obtain a system of equations from which the possible values of the constants a and Ω can be determined. The answer may also include ε, ω.

First method

Note that it is enough to consider one of the solutions, since the other is obtained by rotating the system relative to the z axis; therefore, by symmetry, the dependence coefficients will not change. Let us substitute the first solution option into the answer from the previous section.

$$\begin{pmatrix} x \\ y \end{pmatrix} = A e^{at} \begin{pmatrix} \cos(\Omega t) \\ \sin(\Omega t) \end{pmatrix}$$

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = Ae^{at} \begin{pmatrix} a\cos(\Omega t) - \Omega\sin(\Omega t) \\ a\sin(\Omega t) + \Omega\cos(\Omega t) \end{pmatrix}$$

$$\begin{pmatrix} \ddot{x} \\ \ddot{y} \end{pmatrix} = Ae^{at} \begin{pmatrix} \left(a^2 - \Omega^2\right)\cos(\Omega t) - 2a\Omega\sin(\Omega t) \\ \left(a^2 - \Omega^2\right)\sin(\Omega t) + 2a\Omega\cos(\Omega t) \end{pmatrix}$$

Substituting into the answer from the previous part:

$$\begin{pmatrix} \left(a^2 - \Omega^2\right)\cos(\Omega t) - 2a\Omega\sin(\Omega t) \\ \left(a^2 - \Omega^2\right)\sin(\Omega t) + 2a\Omega\cos(\Omega t) \end{pmatrix} = \begin{pmatrix} -\varepsilon\omega\sin(\Omega t) \\ \varepsilon\omega\cos(\Omega t) \end{pmatrix} - \begin{pmatrix} a\varepsilon\cos(\Omega t) - \varepsilon\Omega\sin(\Omega t) \\ a\varepsilon\sin(\Omega t) + \varepsilon\Omega\cos(\Omega t) \end{pmatrix}$$

Equating the coefficients of the sine and cosine terms, we obtain a system of equations for a and Ω:

$$\begin{cases} a^2 - \Omega^2 = -a\varepsilon \\ 2a\Omega = \varepsilon\omega - \varepsilon\Omega \end{cases}$$

Second method

Let us consider planar motion in polar coordinates (r, φ).

$$\dot{\vec{r}} = \dot{r}\vec{e}_r + \dot{\varphi}r\vec{e}_\varphi$$

$$\ddot{\vec{r}} = \left(\ddot{r} - \dot{\varphi}^2 r\right)\vec{e}_r + \left(\ddot{\varphi}r + 2\dot{r}\dot{\varphi}\right)\vec{e}_\varphi$$

Writing Newton's second law in projection onto $\vec{e}_r$ and $\vec{e}_\varphi$:

$$\begin{cases} \ddot{r} - \dot{\varphi}^2 r = -\varepsilon\dot{r} \\ \ddot{\varphi}r + 2\dot{r}\dot{\varphi} = \varepsilon\left(\omega r - \dot{\varphi}r\right) \end{cases}$$

Substituting $r = Ae^{at}$ and $\varphi = \Omega t$:

$$\begin{cases} a^2 - \Omega^2 = -\varepsilon a \\ 2a\Omega = \varepsilon\omega - \varepsilon\Omega \end{cases}$$

We obtain the same system of equations as in the first method.

Answer:

$$\begin{cases} a^2 - \Omega^2 = -\varepsilon a \\ 2a\Omega = \varepsilon\omega - \varepsilon\Omega \end{cases}$$

B4 — 1.00 pts

Solve the resulting system of equations and find suitable pairs of values (a_1, Ω_1) and (a_2, Ω_2) if $a_2 > a_1$. Express your answer using ε, ω.

To solve, we transform the equations:

$$\begin{cases} \Omega^2 = \left(a + \frac{\varepsilon}{2}\right)^2 - \frac{\varepsilon^2}{4} \\ \Omega = \frac{\varepsilon\omega}{2\left(a + \frac{\varepsilon}{2}\right)} \end{cases}$$

Let $\gamma = \left(a + \frac{\varepsilon}{2}\right)^2$.

$$\frac{\varepsilon^2\omega^2}{4\gamma} = \Omega^2 = \gamma - \frac{\varepsilon^2}{4}$$

$$\gamma^2 - \frac{\varepsilon^2}{4}\gamma = \frac{\varepsilon^2\omega^2}{4} = \left(\gamma - \frac{\varepsilon^2}{8}\right)^2 - \frac{\varepsilon^4}{64}$$

$$\gamma = \frac{\varepsilon^2}{8} \pm \sqrt{\frac{\varepsilon^2\omega^2}{4} + \frac{\varepsilon^4}{64}}$$

Note that $\gamma > 0$, therefore only the root with the $+$ sign is suitable.

$$a + \frac{\varepsilon}{2} = \pm\sqrt{\frac{\varepsilon^2}{8} + \sqrt{\frac{\varepsilon^2\omega^2}{4} + \frac{\varepsilon^4}{64}}}$$

$$a = \varepsilon\left(-\frac{1}{2} \pm \sqrt{\frac{1}{8} + \sqrt{\frac{\omega^2}{4\varepsilon^2} + \frac{1}{64}}}\right)$$

$$\Omega = \pm\frac{\omega}{\sqrt{\frac{1}{2} + \sqrt{\frac{4\omega^2}{\varepsilon^2} + \frac{1}{4}}}}$$

Answer:

$$a_1 = \varepsilon\left(-\frac{1}{2} - \sqrt{\frac{1}{8} + \sqrt{\frac{\omega^2}{4\varepsilon^2} + \frac{1}{64}}}\right)$$

$$a_2 = \varepsilon\left(-\frac{1}{2} + \sqrt{\frac{1}{8} + \sqrt{\frac{\omega^2}{4\varepsilon^2} + \frac{1}{64}}}\right)$$

$$\Omega_1 = -\frac{\omega}{\sqrt{\frac{1}{2} + \sqrt{\frac{4\omega^2}{\varepsilon^2} + \frac{1}{4}}}}$$

$$\Omega_2 = \frac{\omega}{\sqrt{\frac{1}{2} + \sqrt{\frac{4\omega^2}{\varepsilon^2} + \frac{1}{4}}}}$$

B5 — 0.50 pts

Find approximate values of these quantities in the cases $\varepsilon \ll \omega$ and $\varepsilon \gg \omega$.

Answer: For $\varepsilon \ll \omega$:

$$a_1 = -\sqrt{\frac{\omega\varepsilon}{2}} = \Omega_1$$

$$a_2 = \sqrt{\frac{\omega\varepsilon}{2}} = \Omega_2$$

For $\varepsilon \gg \omega$:

$$a_1 = -\varepsilon$$

$$a_2 = 0$$

$$\Omega_1 = -\omega$$

$$\Omega_2 = \omega$$

B6 — 1.50 pts

An arbitrary solution to the system of equations has the form

$$(x(t), y(t)) = A\vec{r}_{11} + B\vec{r}_{12} + C\vec{r}_{21} + D\vec{r}_{22}.$$

Using the initial conditions given in the introduction to this part, express A, B, C, and D in terms of r_0, a_1, a_2, Ω_1, and Ω_2.

Initial conditions: $x(0) = 0$, $y(0) = r_0$, $\dot{x}(0) = 0$ and $\dot{y}(0) = 0$.

$$\begin{cases} A + B = r_0 \\ C + D = 0 \\ a_1 A + a_2 B - \Omega_1 C - \Omega_2 D = 0 \\ a_1 C + a_2 D + \Omega_1 A + \Omega_2 B = 0 \end{cases}$$

$$D = -C$$

$$\begin{cases} A + B = r_0 \\ a_1 A + a_2 B = C\left(\Omega_1 - \Omega_2\right) \\ \Omega_1 A + \Omega_2 B = C\left(a_2 - a_1\right) \end{cases}$$

$$\frac{a_1 A + a_2 B}{\Omega_1 A + \Omega_2 B} = \frac{\Omega_1 - \Omega_2}{a_2 - a_1}$$

$$A\left(a_1\left(a_2 - a_1\right) + \Omega_1\left(\Omega_2 - \Omega_1\right)\right) = B\left(a_2\left(a_1 - a_2\right) + \Omega_2\left(\Omega_1 - \Omega_2\right)\right)$$

$$r_0 = A\frac{a_2\left(a_1 - a_2\right) + \Omega_2\left(\Omega_1 - \Omega_2\right) + a_1\left(a_2 - a_1\right) + \Omega_1\left(\Omega_2 - \Omega_1\right)}{a_2\left(a_1 - a_2\right) + \Omega_2\left(\Omega_1 - \Omega_2\right)}$$

$$r_0 = A\frac{\left(a_2 - a_1\right)^2 + \left(\Omega_2 - \Omega_1\right)^2}{a_2\left(a_2 - a_1\right) + \Omega_2\left(\Omega_2 - \Omega_1\right)}$$

$$A = r_0\frac{a_2\left(a_2 - a_1\right) + \Omega_2\left(\Omega_2 - \Omega_1\right)}{\left(a_2 - a_1\right)^2 + \left(\Omega_2 - \Omega_1\right)^2}$$

$$B = -r_0\frac{a_1\left(a_2 - a_1\right) + \Omega_1\left(\Omega_2 - \Omega_1\right)}{\left(a_2 - a_1\right)^2 + \left(\Omega_2 - \Omega_1\right)^2}$$

$$C = \frac{a_1 A + a_2 B}{\Omega_1 - \Omega_2} = r_0\frac{a_2\Omega_1 - a_1\Omega_2}{\left(a_2 - a_1\right)^2 + \left(\Omega_2 - \Omega_1\right)^2}$$

$$D = r_0\frac{a_1\Omega_2 - a_2\Omega_1}{\left(a_2 - a_1\right)^2 + \left(\Omega_2 - \Omega_1\right)^2}$$

Answer:

$$A = r_0\frac{a_2\left(a_2 - a_1\right) + \Omega_2\left(\Omega_2 - \Omega_1\right)}{\left(a_2 - a_1\right)^2 + \left(\Omega_2 - \Omega_1\right)^2}$$

$$B = -r_0\frac{a_1\left(a_2 - a_1\right) + \Omega_1\left(\Omega_2 - \Omega_1\right)}{\left(a_2 - a_1\right)^2 + \left(\Omega_2 - \Omega_1\right)^2}$$

$$C = r_0\frac{a_2\Omega_1 - a_1\Omega_2}{\left(a_2 - a_1\right)^2 + \left(\Omega_2 - \Omega_1\right)^2}$$

$$D = r_0\frac{a_1\Omega_2 - a_2\Omega_1}{\left(a_2 - a_1\right)^2 + \left(\Omega_2 - \Omega_1\right)^2}$$

B7 0.50 pts

Draw qualitatively the trajectory of the particle in the xy plane for cases $\varepsilon \ll \omega$ and $\varepsilon \gg \omega$.

For $\varepsilon \ll \omega$ we get:

$$x(t) = r_0 \cos\Omega_2 t \cosh\Omega_2 t$$

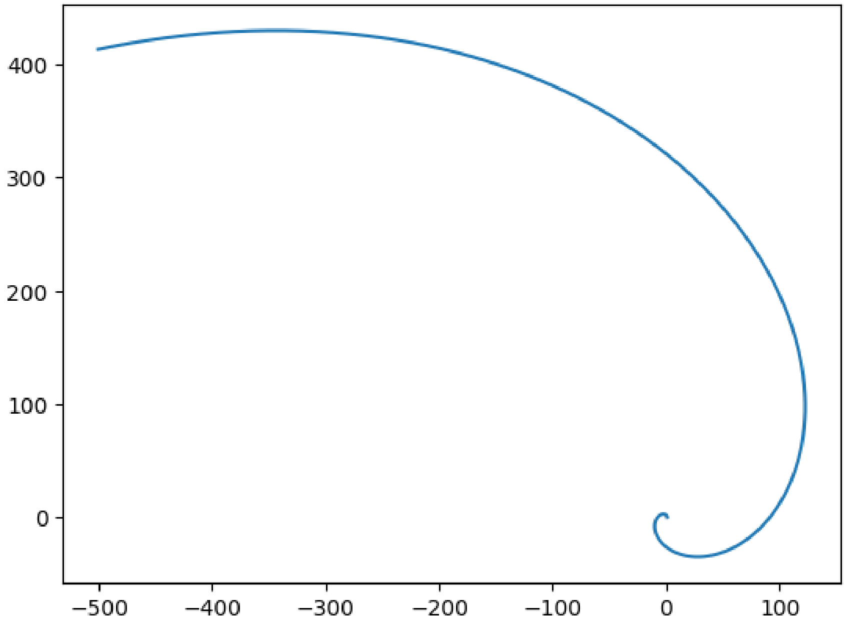
$$y(t) = r_0 \sin\Omega_2 t \sinh\Omega_2 t$$

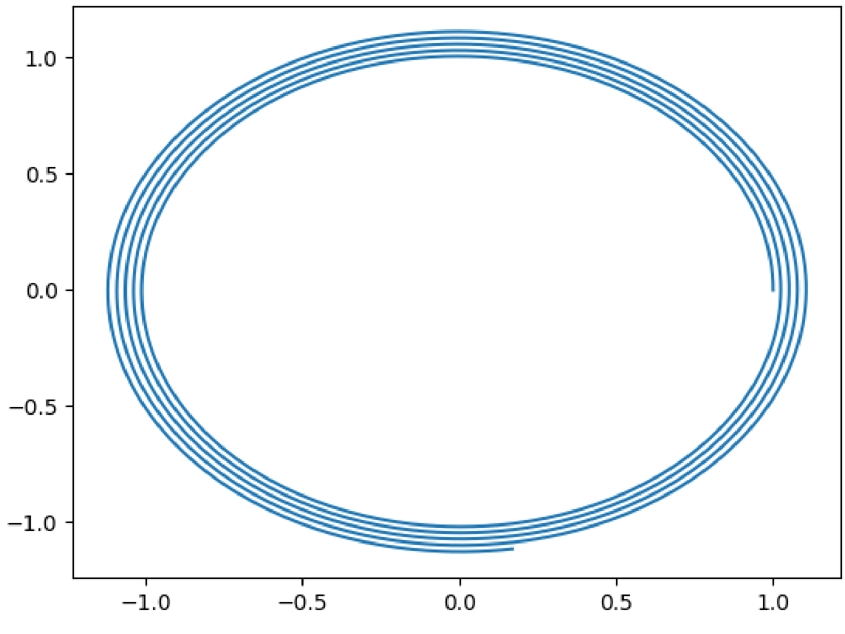
If we neglect the decaying exponential, we obtain a logarithmic spiral. The angle between the radius vector and the velocity a long time after the start of motion is equal to 45°.

At $\varepsilon \gg \omega$ the particle moves along a circle of radius r_0 with a slowly increasing radius:

$$x(t) \approx r_0 \cos\omega t$$

$$y(t) \approx r_0 \sin \omega t$$





B8 **1.00 pts**

Find the energy loss power P, which goes into heat, at the moment when the particle is at a point with coordinates $\vec{r} = (x, y, z)$ and moves with a velocity $\vec{v} = (v_x, v_y, v_z)$.

Let us write down the law of energy change for the system:

$$m\left(\vec{v}, \dot{\vec{v}}\right) = m\left(\vec{g}, \vec{v}\right) - P + W,$$

where W is the power of the external force acting on the medium. Let us find W. For the medium to rotate with a constant angular velocity, a force with a zero projection onto the velocity direction must act on each of its sections. Therefore, on the region where the particle is located, a force $-\vec{F}$ acts from the particle, along with an external force $\vec{F}$.

$$W = \left(\vec{F}, [\vec{\omega}, \vec{r}]\right)$$

$$\left(\vec{v}, m\vec{g} + \vec{F}\right) = m\left(\vec{g}, \vec{v}\right) - P + \left(\vec{F}, [\vec{\omega}, \vec{r}]\right)$$

$$P = \left(\vec{F}, [\vec{\omega}, \vec{r}] - \vec{v}\right) = \varepsilon m v_{\mathrm{rel}}^2 = \varepsilon m\left(v_z^2 + (v_x + \omega y)^2 + (v_y - \omega x)^2\right)$$

Answer:

$$P = \left(\vec{F}, [\vec{\omega}, \vec{r}] - \vec{v}\right) = \varepsilon m v_{\mathrm{rel}}^2 = \varepsilon m\left(v_z^2 + (v_x + \omega y)^2 + (v_y - \omega x)^2\right)$$